Exercises: Concentration Inequalities, LLN, CLT & Convergence Mathematical Foundations for Machine Learning

$P(X \geq a) = E[\mathbf{1}_{X \geq a}]$ -> $P(X \geq a) \leq \frac{E[h(X)]}{h(a)}$

A random variable will not be many times much bigger than it's expected value -> else it's expected value would be bigger

Concentration Inequalities

Exercise 1.1 [Easy] $Z = |X - \mu|$ abs. deviance from mean $h(z) = z^2 \mathbf{1}_{z \geq 0}$ -> $P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}$

Let X be a random variable with $\mathbb{E}[X] = 5$ and $\mathrm{Var}(X) = 4$.

a) Using Chebyshev's inequality, find an upper bound for $P(|X - 5| \geq 3)$.

b) Using Chebyshev's inequality, find a value k such that $P(|X - 5| \geq k) \leq 0.1$.

Exercise 1.2 [Easy]

Let $X_1, \ldots, X_n$ be i.i.d. random variables with $X_i \in [0, 1]$ and $\mathbb{E}[X_i] = \mu$. Using Hoeffding's inequality, find the smallest n such that

$$P\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right| \geq 0.05\right) \leq 0.02.$$

Take h(x) = e^sx, and X = $\frac{1}{n}\sum_{i=1}^{n}(X_i - E[X_i])$ mult n conviniently, apply Markov + hoeff lemma, then minimize over s and so:

$P\left(\frac{1}{n}\sum(X_i - E[X_i]) > t\right) \leq e^{-\frac{2n^2t^2}{\sum_{i=1}^{n}(b_i - a_i)^2}}$

Exercise 1.3 [Medium]

it bounds the sampled mean from the real mean difference

Let $X \sim \mathcal{N}(0, \sigma^2 I_d)$ be a d-dimensional Gaussian random vector and let $Z = \|X\|^2$.

a) Compute $\mathbb{E}[Z]$ and $\mathrm{Var}(Z)$.

b) Using Chebyshev's inequality, show that

$\mathrm{Var}\left(\sum_{i=1}^{d} X_i^2\right) = \sum_{i=1}^{d} \mathrm{Var}(X_i^2)$ · because independent

$\lim_{n\to\infty} \mathbb{E}(|X_n - X|^p) = 0.$

$$P\left(\left|\|X\|^2 - d\sigma^2\right| > t\sqrt{d}\,\sigma^2\right) \leq \frac{2}{t^2}.$$

with expected value we care about magnitude

also, with Xn-X >= e is just yes or no. The p lets us set the sensibility.

c) What does this tell us about the distribution of $\|X\|$ as $d \to \infty$? Express your answer in terms of where the mass of X concentrates.

$\mathrm{P}(\omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)) = 1.$

seq. we defined with rand. var. inside: x/n

$\lim_{n\to\infty} \mathrm{P}(|X_n - X| \geq \varepsilon) = 0.$

if seq. conv. dist. X with more than e prob 0

Exercise 1.4 [Medium]

Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli random variables with $P(X_i = 1) = p$ and $P(X_i = 0) = 1 - p$, and let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.

a) Derive upper bounds on $P(|\bar{X}_n - p| \geq \varepsilon)$ using both Chebyshev's and Hoeffding's inequality.

b) For $\varepsilon = 0.1$ and $p = 0.5$, determine numerically for which values of n the Hoeffding bound is tighter than the Chebyshev bound.

Exercise 1.5 [Hard]

Let $X \sim \mathcal{N}(0, \sigma^2 I_d)$ and define $Z = \|X\|^2$.

a) Show that $\mathbb{E}[\langle X, AX\rangle] = \sigma^2 \mathrm{tr}(A)$ for any symmetric matrix $A \in \mathbb{R}^{d\times d}$.

b) Let $B \in \mathbb{R}^{m\times d}$. Using part (a), show that $\mathbb{E}[\|BX\|^2] = \sigma^2\|B\|_F^2$.

c) Using Chebyshev's inequality and part (b), derive a bound on

$$P\left(\left|\frac{1}{d}\|BX\|^2 - \frac{\sigma^2\|B\|_F^2}{d}\right| > \varepsilon\right)$$

in terms of d, σ, $\|B\|_F$, $\|B\|_{\mathrm{op}}$ and ε. You may use $\mathrm{Var}(\|BX\|^2) \leq 2\sigma^4\|B\|_F^2\|B\|_{\mathrm{op}}^2$ without proof.

Law of Large Numbers & Central Limit Theorem

Exercise 2.1 [Easy]

Let $X_1, X_2, \ldots$ be i.i.d. random variables with $\mathbb{E}[X_i] = 3$ and $\mathrm{Var}(X_i) = 12$.

a) What does the Law of Large Numbers tell us about $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ as $n \to \infty$?

b) Using the Central Limit Theorem, state the approximate distribution of $\bar{X}_n$ for large n.

c) Approximate $P(\bar{X}_{100} > 3.5)$ using the CLT. You may use $\Phi(1.44) \approx 0.925$, where Φ is the standard normal CDF.

Exercise 2.2 [Medium]

A biased die has probability distribution: $P(X = k) = \frac{k}{21}$ for $k = 1, 2, \ldots, 6$.

a) Compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

b) If the die is rolled $n = 200$ times, use the CLT to approximate the probability that the sample mean exceeds 4.5. You may use $\Phi(1.61) \approx 0.946$.

c) Using Chebyshev's inequality, find the minimum number of rolls n needed to guarantee that $P(|\bar{X}_n - \mathbb{E}[X]| \geq 0.5) \leq 0.05$.

Exercise 2.3 [Hard]

Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Define $S_n = \sum_{i=1}^{n} X_i$.

a) Show using Chebyshev's inequality that $\bar{X}_n = \frac{S_n}{n}$ converges to μ in probability (this is the Weak Law of Large Numbers).

b) Let $g : \mathbb{R} \to \mathbb{R}$ be a continuous function. Using the CLT, show that for large n:

$$\mathbb{E}[g(\bar{X}_n)] \approx g(\mu) + \frac{\sigma^2}{2n} g''(\mu).$$

Hint: Use a second-order Taylor expansion of g around μ and the CLT approximation.

c) Apply part (b) to estimate $\mathbb{E}[e^{\bar{X}_n}]$ when $\mu = 0$ and $\sigma^2 = 1$, and compare it to the exact value e^{μ} as $n \to \infty$.

Convergence of Random Variables

Exercise 3.1 [Easy]

State whether each of the following implications is true or false. Give a brief justification.

a) Convergence in probability implies convergence in distribution.

b) Convergence in distribution implies convergence in probability.

c) If $X_n \xrightarrow{P} c$ for a constant c, does $X_n \xrightarrow{d} c$?

Exercise 3.2 [Medium]

Let $U_1, U_2, \ldots$ be i.i.d. Uniform$(0, 1)$ random variables and define $M_n = \max(U_1, \ldots, U_n)$.

a) Show that $P(M_n \leq x) = x^n$ for $x \in [0, 1]$.

b) Determine whether M_n converges in probability to some limit. If so, identify the limit.

c) Find the distribution of $Y_n = n(1 - M_n)$ as $n \to \infty$. What type of convergence does this represent?

Exercise 3.3 [Hard]

Let $X_1, X_2, \ldots$ be i.i.d. Uniform$(0, 1)$ and define

$$Y_n = \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \frac{1}{3}\right).$$

a) Compute $\mathbb{E}[X_i^2]$ and $\mathrm{Var}(X_i^2)$.
b) Using the CLT, determine the asymptotic distribution of Y_n.
c) Use part (b) to approximate $P(Y_{400} > 0.05)$. You may use $\Phi(0.47) \approx 0.681$.
d) Is it possible to apply Hoeffding's inequality to bound $P\big(\big|\frac{1}{n}\sum_{i=1}^n X_i^2 - \frac{1}{3}\big| \geq t\big)$? If so, state the bound.

Confidence Intervals for the Test Error

Exercise 4.1 [Easy]

A classifier f is evaluated on $n = 1000$ i.i.d. test samples and achieves an empirical test error of $\widehat{\mathrm{err}}_f = 0.08$.

a) Using Hoeffding's inequality, construct a 95% confidence interval for the true error err_f.
b) How many test samples are needed to guarantee the confidence interval has width at most 0.02 at 95% confidence?

Exercise 4.2 [Medium]

Let f be a binary classifier with true error $\mathrm{err}_f = P(f(X) \neq Y)$, evaluated on n i.i.d. test samples.

a) Write $\widehat{\mathrm{err}}_f$ in terms of the indicator variables $\mathbf{1}_{f(X_i)\neq Y_i}$ and show that $\mathrm{Var}(\widehat{\mathrm{err}}_f) = \frac{\mathrm{err}_f(1-\mathrm{err}_f)}{n}$.
b) Using the fact that $\mathrm{err}_f(1 - \mathrm{err}_f) \leq \frac{1}{4}$, derive a confidence interval via Chebyshev's inequality.
c) For $n = 10{,}000$ and $\varepsilon = 0.01$, compare the upper bounds from Chebyshev and Hoeffding on $P(|\widehat{\mathrm{err}}_f - \mathrm{err}_f| \geq \varepsilon)$. Which one is tighter?

Exercise 4.3 [Hard]

You are evaluating two classifiers f and g on the same test set of n i.i.d. samples. Define $\Delta = \mathrm{err}_f - \mathrm{err}_g$ and $\hat{\Delta} = \widehat{\mathrm{err}}_f - \widehat{\mathrm{err}}_g$.

a) Let $Z_i = \mathbf{1}_{f(X_i)\neq Y_i} - \mathbf{1}_{g(X_i)\neq Y_i}$. Express $\hat{\Delta}$ in terms of Z_i and note that $Z_i \in [-1, 1]$.
b) Using Hoeffding's inequality on Z_i, derive a confidence interval for Δ.
c) If $n = 5000$, $\widehat{\mathrm{err}}_f = 0.12$ and $\widehat{\mathrm{err}}_g = 0.10$, determine with 95% confidence whether there is a statistically significant difference between the two classifiers.

Mixed Challenging Exercises

Exercise 5.1 [Hard]

A machine learning engineer trains a model and wants to evaluate it rigorously. The test set has n i.i.d. samples, each in $\mathbb{R}^d$, with labels in $\{0, 1\}$.

a) The model achieves $\widehat{\mathrm{err}}_f = 0.03$ on $n = 500$ test samples. Using Hoeffding's inequality, compute a 99% confidence interval for err_f.
b) The engineer claims: "My model has at most 5% error." Using part (a), can this claim be supported at 99% confidence?
c) How large must n be so that a 99% confidence interval using Hoeffding has half-width at most 0.01?
d) Using the CLT, approximate the distribution of $\widehat{\mathrm{err}}_f$ for $n = 500$ assuming $\mathrm{err}_f = 0.03$. Compare the CLT-based 99% confidence interval to the one from Hoeffding. You may use $z_{0.005} = 2.576$.

Exercise 5.2 [Hard]

Consider a dataset of n i.i.d. samples $X_1, \ldots, X_n$ drawn uniformly from $[0,1]^d$.

a) For a fixed point $p \in [0,1]^d$, compute $\mathbb{E}[\|X_i - p\|^2]$ and show that it does not depend on p when $p = (\frac{1}{2}, \ldots, \frac{1}{2})$. *Hint:* Use $\mathrm{Var}(X_{i,j}) = \frac{1}{12}$ for $X_{i,j} \sim \mathrm{Uniform}(0,1)$.

b) Using Hoeffding's inequality applied coordinate-wise, show that

$$P\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_{i,j} - \frac{1}{2}\right| \geq t\right) \leq 2e^{-2nt^2}$$

for each coordinate j.

c) Using a union bound and part (b), find a value t such that with probability $\geq 1-\delta$, the sample mean $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ satisfies $\|\bar{X}_n - (\frac{1}{2}, \ldots, \frac{1}{2})\|_\infty \leq t$.

d) Interpret the result: how does the number of samples n needed to achieve $\|\bar{X}_n - \mu\|_\infty \leq \varepsilon$ with confidence $1-\delta$ scale with the dimension d?